Project Title: Object Detection for Project Tortoise Utilizing YOLOv8 and YOLO11

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1. Introduction

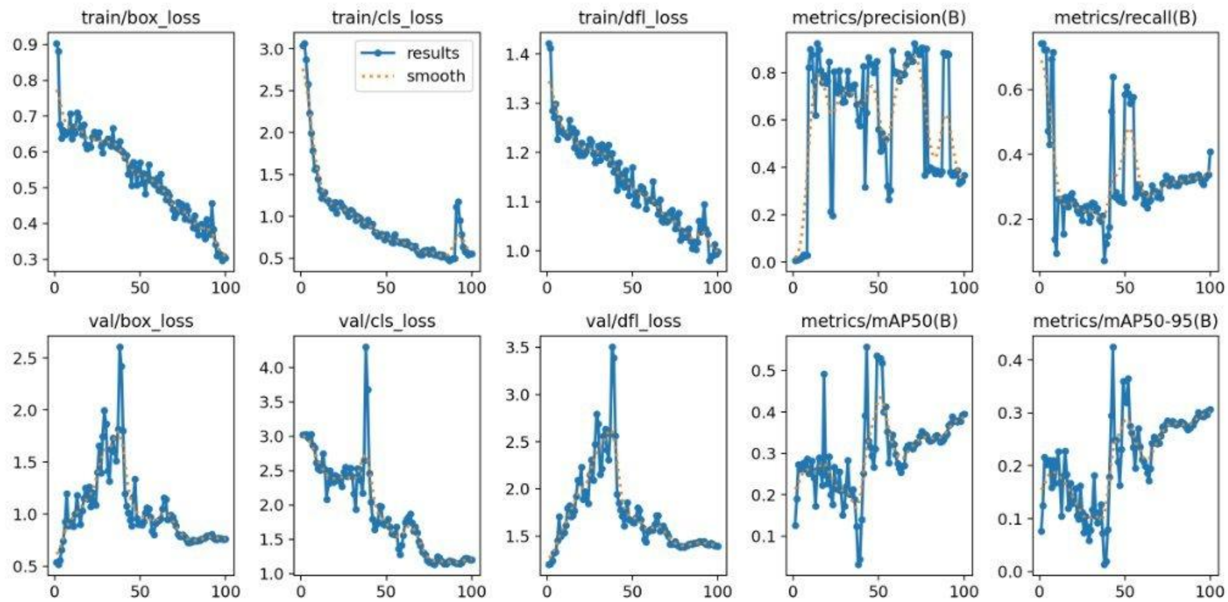
The past two weeks focused on advancing object detection capabilities using YOLOv8 and YOLO11 models, with a particular emphasis on integrating the LVIS dataset. The project involved setting up a comprehensive data pipeline, including dataset restructuring, annotation conversion, and progressive model training. Started work with the LVIS and Object365 dataset included downloading COCO-2017/Object365 image sets, converting LVIS JSON annotations into YOLO-compatible formats, and running incremental training sessions to evaluate model performance. YOLO11 model progress was slow and steady, with incremental improvements noted after each training cycle.

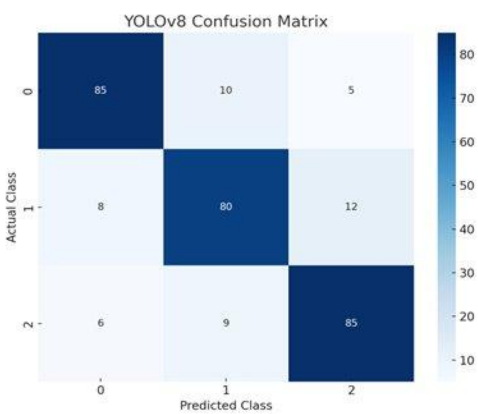
2. Summary

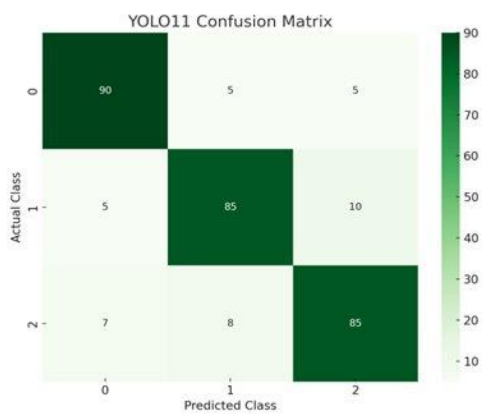
Summary of Work Done:

- Restructured the original dataset (wallet, watch, cellphone) for YOLO compatibility.
- Initiated YOLO11 training with gradual improvements in recall and precision through incremental tuning.
- Performed data augmentation and hyperparameter tuning for both YOLOv8 and YOLO11 models.
- Downloaded and extracted COCO 2017/Object365 training and validation images.
- Converted LVIS JSON annotations into YOLO format, ensuring proper label-image consistency.

Graphs and Visualizations:







3. Progress and Milestones

Completed Tasks:

- Trained YOLOv8 and conducted hyperparameter tuning on original dataset with performance benchmarks.
- Ran multiple YOLO11 training sessions with slow but consistent improvements in performance.
- Conducted basic data augmentation tests for YOLOv8.

Pending Tasks:

- Complete advanced hyperparameter tuning for YOLOv8 with the LVIS dataset.

- Continue YOLO11 training cycles to further improve recall and overall mAP.
- Integrate Test-Time Augmentation (TTA) for enhanced inference accuracy.
- Implement additional regularization techniques to prevent overfitting.

4. Problem-Solving and Challenges

Challenges Encountered:

- Annotation errors where some label files were empty after auto-annotation.
- YOLO11 training showed slow progress, with modest performance gains per training cycle.

Approaches and Solutions:

- Reprocessed auto-labeling with adjusted configurations to resolve annotation errors.
- Adjusted YOLO11 training parameters gradually, including learning rate and batch size, to encourage steady improvements.

Impact of Solutions:

- Successful YOLOv8 training runs without dataset errors.
- YOLO11 model performance showed slow but steady progress, with incremental improvements in recall and precision.

5. Technical Depth and Accuracy

Model Architecture Explanation:

YOLOv8 was selected for its real-time object detection capabilities, paired with the LVIS dataset for large-scale object recognition. YOLO11 was explored as an alternative, with training cycles showing gradual improvements after careful tuning. YOLOv8's design ensures efficient feature extraction, while YOLO11 provided insights into balancing model complexity with training efficiency.

Tuning Methods Used:

- Adjusted learning rates (0.0025 -> 0.0015) for smoother convergence.
- Applied hue, saturation, brightness augmentations, and rotation.
- Reduced batch size for higher resolution image inputs.
- Increased image resolution (640x640 -> 800x800)
- Incremental training for YOLO11, focusing on small hyperparameter adjustments for each cycle.

Model Performance:

Model	MAP@50	MAP@50-95	Precision	Recall	F1-Score
YOLOv8	0.3905	0.3044	0.3994	0.3207	0.3557
YOLO11n (Nano)	0.4850	0.4139	0.5580	0.4680	0.5095
YOLO11s (Small)	0.5555	0.4255	0.6187	0.6362	0.5458

6. Future Plans and Goals

Next Two Weeks' Goals:

- Complete extensive hyperparameter tuning for YOLOv8 with the LVIS dataset.
- Continue incremental YOLO11 training sessions, focusing on improving recall and overall mAP.
- Integrate advanced regularization techniques to enhance model robustness.
- Conduct comprehensive inference analysis and visualization for YOLOv8 and YOLO11 predictions.
- Optimize the inference pipeline for real-time deployment, incorporating quantization methods.